Name: _____

TOPIC TEST 2

Chapter 8: Reducing balance loans and annuities

- Time allowed: 45 minutes
- CAS calculators may be used.
- Part A: 5 multiple-choice questions (5 marks)
- Part B: 5 short-answer questions (15 marks)
- Part C: 1 analysis question (18 marks)

Part A

Multiple-choice questions

5 multiple-choice questions
1 mark each: 5 marks
Circle the correct answer.

1 Emma borrows $40 000 from the bank to finance an overseas trip. Interest is charged on this loan at a rate of 5.5% p.a., calculated monthly. If Emma pays $600 a month towards the loan, how much does she owe the bank after 5 years?

A $11 299.66

B $4000

C $93 956.64

D $16 278.40

E $957 590.82

[1 mark]

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2 Luke takes out a loan for $250 000 at 5.15% p.a., calculated monthly, to be paid back over a period of 25 years. If, after 15 years, Luke increases his monthly repayments to $2000, how many months earlier will he pay off the loan?

A 143

B 83

C 263

D 217

E 37

[1 mark]

3 Maddison receives an inheritance of $0.75 million. She invests the money at 3.99% p.a., compounding monthly. If Maddison chooses to receive monthly payments of $5000, how long will the money last (to the nearest month)?

A 20 years 8 months

B 17 years 3 months

C 12 years 6 months

D 17 years 4 months

E 8 years 8 months

[1 mark]

4 Hamish receives an inheritance of $300 000 and invests in a perpetuity that pays him 4.75% interest p.a., compounding monthly. How much can Hamish expect to receive each month from this perpetuity?

A $63 157.89

B $14 250

C $1187.50

D $631.58

E $142.50

[1 mark]

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5 Amina opens a savings account with an interest rate of 4.15%. She begins with a $10 000 investment and deposits $1500 per month. What is the amount of interest earned over 8 years?

A $184 367.77

B $30 367.77

C $156 508.67

D $20 367.77

E $2508.67

[1 mark]

Part B

Short-answer questions

5 short-answer questions
15 marks
Show your working where appropriate.

1 Dominic took out a loan for $28 000. He makes monthly repayments of $500 and interest is calculated at a rate of 6.2% p.a., adjusted monthly.

a Write a difference equation that gives the amount still owing on the loan after n months.

b How much does Dominic owe after 4 years?

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c How much interest has Dominic paid in this time?

d At this rate, how long until Dominic pays off the loan?

[4 marks]

2 Natalia borrows $60 000 from a bank at 7.99% p.a. interest, adjusted monthly. If the loan is to be repaid after 10 years, find the:

a monthly repayment

b amount of interest paid.

[2 marks]

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3 Will takes out a loan for $10 000 at 10.75% p.a. interest, adjusted quarterly.

a How many years until the loan is repaid if Will repays $607.99 per quarter?

b If Will needs to pay off the loan in 3 years, find the quarterly repayment required.

c How much interest would Will save by repaying according to part **b** rather than part **a**?

[4 marks]

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4 Michael purchases a $220 000 annuity. Interest is paid at 3.85% p.a. compounded quarterly.

a If he wants the annuity to last 15 years, what would be his quarterly payments from the bank?

b If Michael decides to receive payments of $8000 per quarter, how long would the annuity last?

[2 marks]

5 Rachael wants to set up a perpetuity. She decides to invest in bonds that return 5.99% p.a., compounded monthly.

a If she invests $150 000, how much can she expect to receive each month?

b If Rachael wants to receive $1200 each month, how much more will she need to invest?

[3 marks]

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Part C

Analysis question

1 analysis question
18 marks
Show your working where appropriate.

1 Tess and Stuart are to share equally in an inheritance of $750 000. Tess decides to invest her money, whereas Stuart uses it as a deposit for a house loan.

Tess invests her money in an account at 3.99% p.a., compounding monthly.

a If she leaves this untouched (no deposits or withdrawals) for 10 years, what would be her account balance?

b Tess decides to deposit $200 a month into the account. How much would she have after 10 years?

After 6 years, the interest rate drops to 3.15% p.a.

c How much less money would Tess have after a total of 10 years if she continued depositing $200 a month?

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d To what amount should Tess raise her monthly deposits in order to ensure her bank balance reaches the 10-year total calculated in part **b**?

Once the interest rate dropped, Tess decided that rather than increasing her monthly deposit, she would change her investment to shares earning 4.75% p.a. compounding quarterly and set up a perpetuity.

e How much should Tess expect to receive quarterly from her perpetuity?

f If Tess decided that she would instead prefer to receive quarterly payments of $7500, how long would this annuity now last?

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With his share of the money, Stuart places a deposit on a $600 000 house and takes out a loan for the remainder of the house value.

g If he takes out a 20-year loan at 5.7% p.a. compounding monthly, what monthly repayments would he need to make?

After 5 years, the interest rate dropped to 4.99% p.a.

h If Stuart continued with his monthly repayments, how much:

i time would he save on his loan and

ii how much interest would he save on his loan?

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i If Stuart instead chose to keep the 20-year term, to what amount would his monthly repayments drop?

j Find the total amount of interest Stuart would pay if he chose to keep the 20-year term.

k Would this amount differ from the amount of interest he would have paid if the interest rate had remained the same? If so, by how much?

l What would be your recommendation to Stuart?

[18 marks]

[Total: 38 marks]

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Answers

Part A – Multiple-choice questions [1 mark each]

1 Using:

$N = 5 \times 12 = 60$

$I\% = 5.5$

$PV = 40\,000$

$PMT = -600$

$P/Y = C/Y = 12$

gives an FV of $-\$11\,299.66$

$\therefore$ **A**

2 Using:

$N = 25 \times 12 = 300$

$I\% = 5.15$

$PV = 250\,000$

$P/Y = C/Y = 12$

$FV = 0$

gives a PMT of $-\$1483.41$

Next, using:

$N = 15 \times 12 = 180$

$I\% = 5.15$

$PV = 250\,000$

$P/Y = C/Y = 12$

PMT of $-\$1483.41$

gives an FV of \$138 895.46, which is the amount he still owes after 15 years.

Next, using:

$I\% = 5.15$

$PV = 138\,895.46$

$P/Y = C/Y = 12$

PMT of $-\$2000$

$FV = 0$

gives $N = 82.64$, which is the number of months until he pays off the loan.

83 months (rounded) + 180 months = 263 months in total to pay off the loan.

300 months − 263 months = 37 months saved by increasing payments.

$\therefore$ **E**

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3 Using:

$I\% = 3.99$

$PV = -750\,000$

$PMT = \$5000$

$P/Y = C/Y = 12$

$FV = 0$

gives $N = 208.06$

$\frac{208}{12} = 17.333\ldots$ years, which is 17 years 4 months.

$\therefore$ **D**

4 Using the formula $Q = \frac{Pr}{100}$, where $P = 300\,000$ and $r = \frac{4.75}{12}$, we find $Q = \$1187.50$.

$\therefore$ **C**

5 Using:

$N = 8 \times 12 = 96$

$I\% = 4.15$

$PV = -10\,000$

$PMT = -\$1500$

$P/Y = C/Y = 12$

gives $FV = \$184\,367.77$

$\$184\,367.77 - (10\,000 + 1500 \times 96) = \$30\,367.77$

$\therefore$ **B**

Part B – Short-answer questions

1 a $A_n = \left(1 + \frac{r}{100}\right)A_{n-1} -$ monthly repayment

$r = \frac{6.2}{12}$ (interest rate per month)

Monthly repayment $= 500$

Initial balance $= A_0 = 28\,000$

$A_n = \left(1 + \frac{6.2}{1200}\right)A_{n-1} - 500$, where $A_0 = 28\,000$ **[1 mark]**

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b Using:

$N = 4 \times 12 = 48$

$I\% = 6.2$

$PV = 28\,000$

$PMT = -500$

$P/Y = C/Y = 12$

gives $FV =$ \$8699.07

After 4 years, Dominic owes \$8699.07. **[1 mark]**

c Dominic has paid $\$500 \times 48 = \$24\,000$.

Dominic has paid off $\$28\,000 - \$8699.07 = \$19\,300.93$.

Amount of interest $= \$24\,000 - \$19\,300.93 = \$4699.07$. **[1 mark]**

d Using:

$I\% = 6.2$

$PV = 28\,000$

$PMT = -500$

$FV = 0$

$P/Y = C/Y = 12$

gives $N = 66.28$ months

At this rate, Dominic pays off the loan in 5 years, 6 months. **[1 mark]**

2 a Using:

$N = 10 \times 12 = 120$

$I\% = 7.99$

$PV = 60\,000$

$FV = 0$

$P/Y = C/Y = 12$

gives $PMT = -\$727.65$

The monthly repayment is \$727.65 **[1 mark]**

b Amount of interest paid = total amount repaid − value of loan

$= 727.65 \times 120 - 60\,000 = \$27\,318$.

Natalia paid \$27 318 in interest. **[1 mark]**

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3 a Using:

$I\% = 10.75$

$PV = 10\,000$

$PMT = -\$607.99$

$FV = 0$

$P/Y = C/Y = 4$

gives $N = 22$ quarters

It therefore takes 22 quarters to fully repay the loan, which is $\frac{22}{4} = 5.5$ years. **[1 mark]**

b Using:

$N = 3\ 3\ 4 = 12$

$I\% = 10.75$

$PV = 10\,000$

$FV = 0$

$P/Y = C/Y = 4$

gives $PMT = -\$985.97$

Will would need to pay \$985.97 per quarter to pay off the loan in 4 years. **[1 mark]**

c Interest paid in part **a** = total amount repaid − value of loan

$= 607.99 \times 22 - 10\,000 = \$3375.78.$

Interest paid in part **b** = total amount repaid − value of loan

$= 985.97 \times 12 - 10\,000 = \$1831.64.$

Interest saved = \$3375.78 − \$1831.64 = \$1544.14. **[1 mark]**

Will would save \$1544.14 in interest if he chose to repay according to part **b** rather than part **a**. **[1 mark]**

4 a Using:

$N = 15\ 3\ 4 = 60$

$I\% = 3.85$

$PV = -220\,000$

$FV = 0$

$P/Y = C/Y = 4$

gives $PMT = -\$4843.90$

Michael would receive quarterly payments of \$4843.90. **[1 mark]**

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b $I\% = 3.85$

$PV = -220\,000$

gives $PMT = -\$8000$

$FV = 0$

$P/Y = C/Y = 4$

gives $N = 32.09$ quarters

This is $\frac{32}{4} = 8$ years.

Michael's annuity would last for 8 years. **[1 mark]**

5 a Using the formula $Q = \frac{Pr}{100}$, where $P = 150\,000$ and $r = \frac{5.99}{12}$

gives $Q = \$748.75$.

Rachael would receive \$748.75 per month. **[1 mark]**

b Using the formula $Q = \frac{Pr}{100}$, where $Q = 1200$ and $r = \frac{5.99}{12}$

gives $P = \$240\,400.67$. **[1 mark]**

Rachael will need to invest \$240 400.67 − \$150 000 = \$90 400.67 more in order to receive monthly payments of \$1200. **[1 mark]**

Part C – Analysis question

1 a Compound interest formula: $A = P\left(1 + \frac{r}{100}\right)^n$

$P = 375\,000, r = \frac{3.99}{12}, n = 12 \times 10 = 120$

$A = 375\,000\left(1 + \frac{3.99}{1200}\right)^{120} = \$558\,505.33$

Tess' account balance would be \$558 505.33. **[1 mark]**

b $N = 10 \times 12 = 120$

$I\% = 3.99$

$PV = -375\,000$

$PMT = -\$200$

$P/Y = C/Y = 12$

gives $FV = -\$587\,939.76$

If she deposited \$200 per month, Tess would have \$587 939.76 after 10 years. **[1 mark]**

c $N = 6 \times 12 = 72$

$I\% = 3.99$

$PV = -375\,000$

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$PMT = -\$200$

$P/Y = C/Y = 12$

gives $FV = -\$492\ 482.85$

After 6 years, Tess has \$492 482.85. **[1 mark]**

Next, using:

$N = 4 \times 12 = 48$

$I\% = 3.15$

$PV = -492\ 482.85$

$PMT = -\$200$

$P/Y = C/Y = 12$

gives $FV = -\$568\ 740.19$

After 10 years, Tess has \$568 740.19. **[1 mark]**

As a result of the interest rate drop, Tess would have:

\$587 939.76 − \$568 740.19 = \$19 199.57 less money after 10 years. **[1 mark]**

d After 6 years, Tess has \$492 482.85.

Using:

$N = 4 \times 12 = 48$

$I\% = 3.15$

$PV = -492\ 482.85$

$FV = -587\ 939.76$

$P/Y = C/Y = 12$

gives $PMT = -\$575.84$

Tess should lift her monthly deposits to \$575.84 in order to ensure her bank balance reaches the previous 10-year total (\$587 939.76). **[1 mark]**

e Using the formula $Q = \frac{Pr}{100}$, where $P = 492\ 482.85$ and $r = \frac{4.75}{4}$ gives $Q = \$5848.23$. Therefore, Tess should expect to receive \$5848.23 quarterly from her perpetuity. **[1 mark]**

f Using:

$I\% = 4.75$

$PV = -492\ 482.85$

$PMT = -\$7500$

$FV = 0$

$P/Y = C/Y = 4$

gives $N = 128.17$

Therefore, the annuity would last 128.17 quarters, which is roughly 32 years. **[1 mark]**

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g Loan is for \$600 000 − \$375 000 = \$225 000

$N = 20 \times 12 = 240$

$I\% = 5.7$

$PV = 225\,000$

$FV = 0$

$P/Y = C/Y = 12$

gives $PMT = -1573.27$

Stuart will need to make monthly repayments of \$1573.27. **[1 mark]**

h i $N = 5 \times 12 = 60$

$I\% = 5.7$

$PV = 225\,000$

$PMT = -1573.27$

$P/Y = C/Y = 12$

gives $FV = -190\,069.53$

Stuart will still owe \$190 069.53 after 5 years. **[1 mark]**

Next, using:

$I\% = 4.99$

$PV = 190\,069.53$

$PMT = -1573.27$

$FV = 0$

$P/Y = C/Y = 12$

gives $N = 168.18$ months **[1 mark]**

Time saved = original time − actual time

= 240 months − (60 + 168) months = 12 months.

Therefore, Stuart will save approximately 1 year on his loan. **[1 mark]**

ii Interest saved = original interest to be paid − new interest paid

$= 1573.27 \times 240 - 1573.27 \times (60 + 168)$

$= 377\,584.80 - 358\,705.56$

= \$18 879.24 **[1 mark]**

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i Stuart still owes \$190 069.53 after 5 years.

Using:

$N = 240 - 60 = 180$ **[1 mark]**

$I\% = 4.99$

$PV = 190\,069.53$

$FV = 0$

$P/Y = C/Y = 12$

gives $PMT = -1502.07$

Stuart's monthly repayments would drop to \$1502.07. **[1 mark]**

j Interest paid = total amount repaid − value of loan

$= 1573.27 \times 60 + 1502.07 \times 180 - 225\,000 =$ \$139 768.80. **[1 mark]**

k Original interest paid $= 1573.27 \times 240 - 225\,000 =$ \$152 584.80.

So, yes, this amount differs from the amount of interest Stuart would have paid if the interest rate had remained the same.

Therefore, Stuart saves \$152 584.80 − \$139 768.80 = \$12 816 as a result of the interest rate cut.

[1 mark]

l Stuart is best advised to maintain his monthly payments because he saves 1 year in time, but, importantly, he saves:

\$18 879.24 − \$12 816 = \$6063.24 in interest. **[1 mark]**

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